1. If the standard deviation of a dataset is small, what does this tell you about the data points?

(A) They are widely dispersed.
(B) They are clustered closely around the mean.
(C) The data is skewed.
(D) The data is symmetrical.

2. Consider the small data set from the notes.

6 7 7 7 8 8 9 9 10 11 11

If every value in this data set is increased by 10, what happens to:

- the mean and median and of the set?

(A) Both remain the same.
(B) Both increase by 10.
(C) The mean increases by 10, but the median stays the same.
(D) The mean remains the same, but the median increases by 10.

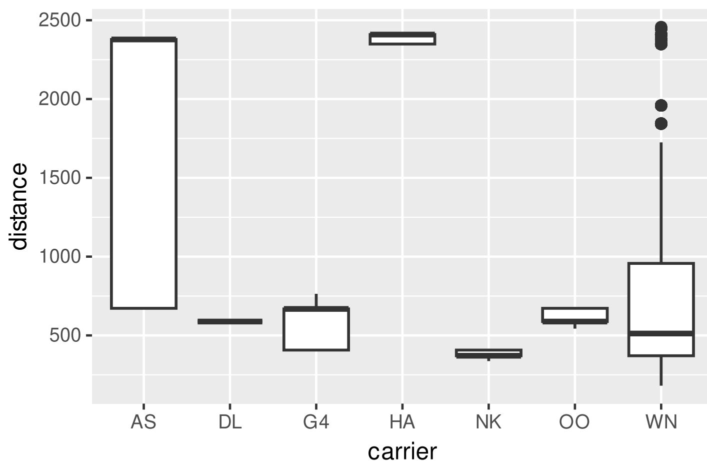
- to the standard deviation and variance of the set?

(A) Both remain the same.
(B) Both increase by 10.
(C) The standard deviation increases by 10, but the variance stays the same.
(D) The variance increases by 10, but the standard deviation stays the same.

3. For which one of the following distributions will the median be a better measure of center than the mean?

(A) Salary data for players in the National Basketball Association (NBA) where most of the players earn the league minimum and a few superstars earn very high salaries in comparison.
(B) Repeated weight measurements of the same 1.6-ounce bag of Peanut M&Ms by students in a large chemistry class using a non-digital balance scale.
(C) Height data from a large random sample of men.
(D) Exam scores with a central peak around an average test score and a few students scoring lower than the average and a few scoring higher.

Below is a smaller version of the data from a future lab called `flights_mini`. It contains all flights out of Oakland (OAK) from December 2020. This data frame is used to create the plot that follows. `distance` refers to the distance a given plane travels on its flight, measured in miles. `carrier` refers to the carrier code for a specific airline.



4. Which of the following interpretations of the plot above are true? *Select all that apply.*

(A) The carrier with the most heavily skewed distance distribution is HA.
(B) The median distance of the flights operated by DL, G4, and OO are roughly equivalent.
(C) The minimum distance traveled in this data set is roughly 200.
(D) The carrier with the greatest variability in distance, as measured by the IQR, is AS.

5. Sketch your best sense of the distribution of the following variable(s). For each, please:

i. Use a form of statistical graphic that emphasizes the important elements of the distribution.
ii. Label the axes and provide plausible values for the tick marks.
iii. Describe in words the shape of the distribution.
iv. State which measure of center and spread would be most appropriate and approximate their values.

Make a note of any assumptions you're making in interpreting these variable names.

How do Berkeley students get to campus?

Scores on a quiz where most do well but a few score very poorly

The `mpg` data frame is available as a part of the `tidyverse` library. It contains information on fuel consumption for 38 models of car between 1999 and 2008. *Datasets can have help files, too*, so be sure to consult them if you need to for the following questions.

6. Write `dplyr` code to isolate just the city miles per gallon and car class variables, and save your result, which should be a two-column data frame, into the object `fuel`.

7. Using the `fuel` data frame, write `dplyr` code to calculate the median and IQR city miles per gallon for the vehicles in the dataset and copy it below. The result of your code should be one data structure.

8. Also using the `fuel` data frame, `dplyr` code to calculate the mean and standard deviation city miles per gallon for the vehicles in the dataset *for each class of car* and copy it below. The result of your code should be one data structure.